

TECHNICAL NOTE

Title:	Conclusion of Lacq Controlled Release Technical Limitations
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Date:	20/10/2024

1. SUMMARY

Flylogix was invited to participate in a controlled release experiment at Lacq between 16th -20th September 2024. Prior to the trial Flylogix noted the facility where the trial was conducted imposed a number of constraints that impacted the quality of the measurement in technical note, FX-TN-034. In the spirit of robust experimentation Flylogix highlighted these constraints prior to commencing the campaign.



Fig 1. Flylogix Aircraft conducting methane emission measurement of offshore asset

This document notes the actual constraints experienced during the trial.

2. DISCUSSION

Flylogix specialises in conducting asset level emission measurements of offshore oil and gas assets using a long range, fixed wing, drone. By conducting the entire operation using small uncrewed aircraft operating from runways on the coast this reduces the cost, risk and environmental impact of measuring offshore methane emissions. Over the last 6 years Flylogix has refined an approach where the aircraft flies a cylindrical pattern around the offshore asset forming a closed cylinder whilst making accurate measurements of the local wind, facilitating a mass flow calculation. Flylogix has optimised stand offs to maximise point concentration whilst minimising the impact of wake and plume instabilities. Flylogix use the mass flux method to successfully measure the total emissions down to approximately 1kg/hr with an uncertainty of 2kg/hr. This approach is robust to upwind contamination, plume instability, multiple sources and weather. The method is repeatable, transparent and auditable.

Following successful measurements using this technique of Neptune and TotalEnergies facilities for EDF, Flylogix was encouraged to participate in the TADI trial. Flylogix welcome peer reviewed controlled releases to improve their approach. During planning the Lacq facility told us that we are not able to conduct our standard technique using the fixed wing uncrewed aircraft and flight patterns deployed offshore due to site restrictions. Flylogix therefore engaged a multirotor drone service provider, Air Control Entech, to fly a reduced diameter cylinder at the Lacq facility. Flylogix believe that the point source and absence of a large superstructure creating a wake mean that we can conduct a meaningful replication of the technique with a smaller diameter cylinder.

During the trial Air Control Entech were able to complete closed circuits but not a cylindrical flight path due to site constraints. This distorts the mass-balance contour integral calculation. Furthermore, mesoscale turbulent fluctuations with timescales comparable to the flight time of the drone can negatively impact flux plane measurements and associated mass-balance calculations at these distances, an issue that is normally mitigated by the large standoff employed in our standard approach. We also noted that several surveys were paused due to personnel entering the site during a survey, exacerbating this. Finally it was not possible to locate wind instrumentation in a suitable, free-stream location resulting in poor wind data – and this was compounded by very low windspeeds and inversions during the trials.

Nonetheless we were able to gather a large volume of valuable data and have shared in both the requested spreadsheet and the full report format. We do have a wealth of experience of performing offshore methane surveys and as such would be very keen to help in defining a more representative test of offshore-monitoring solutions in the future.

3. CONCLUSIONS

Flylogix welcomed the opportunity to participate in the TADI trial.

The constraints on aircraft and flight route at Lacq meant the trial was not representative of Flylogix's proven fixed-wing offshore measurement methodology.

Flylogix provided a meaningful test of the AERIS Strato instrument by employing as close a replica of their standard method as the site and drone allowed.

Long term, Flylogix would welcome the opportunity of defining and participating in a more representative offshore controlled release.